Ebullient Securities Performance Report

Team 67

CONTENTS

I. EBY Performance

After implementing and backtesting our strategies on the EBX and EBY datasets, the metrics achieved were as follows.

A. Backtest Summary

TABLE I. Backtest Performance Metrics

Metric	Value
Initial Capital	$10000
Final Capital	$13475.11
Total PnL with TC	$3475.11
Total Transaction Cost	$1267.75
Final Returns	34.75%
Annualized Returns	31.39%
Sharpe Ratio	2.58
Calmar Ratio	14.40
Maximum Drawdown	-2.18%
No. of Days	279
Winning Days	110
Losing Days	88
Best Day	104
Worst Day	252
Best Day PnL	$1099.12
Worst Day PnL	-$136.31
Total Trades	663
Winning Trades	290
Losing Trades	373
Win Rate	43.74%
Average Winning Trade	$46.74
Average Losing Trade	-$15.22
Avg Hold Period (seconds)	7,130.88
Avg Hold Period (minutes)	118.85

B. Performance Visualizations

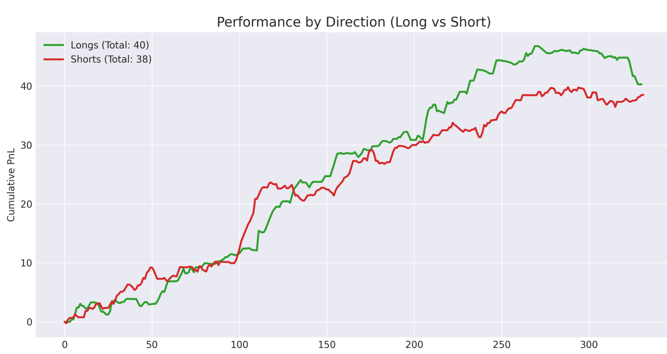

Fig. 1. Performance by Direction (Long vs Short)

1) Performance by Direction The cumulative performance chart shows both long and short positions contributed positively to overall returns, though with different trajectories. From trades 0 to 100, both directions tracked closely together, climbing steadily from 0 to around 10-12 in cumulative PnL with shorts initially showing slight outperformance. From trade 100 onwards, longs began to pull ahead more decisively, accelerating upward to reach a peak of approximately 48 around trade 280, while shorts continued their steady climb but at a more modest pace, reaching about 39 by the end. The divergence became most pronounced after trade 200, where longs captured a strong upward move while shorts plateaued somewhat. In the final 30 trades, longs experienced a notable pullback from their peak while shorts remained relatively stable, suggesting that the late-period deterioration seen in the overall equity curve was primarily driven by long position losses.

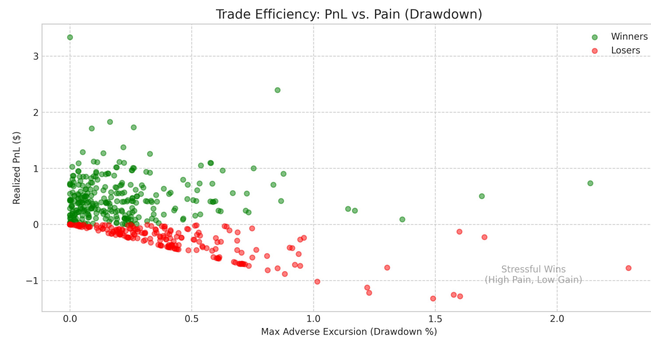

Fig. 2. Trade Efficiency: PnL vs. Pain (Drawdown)

2) Trade Efficiency Analysis The scatter plot reveals a clear efficiency pattern where winning trades cluster in the lower-left quadrant with minimal drawdown (mostly below 0.5%) and realized gains typically between 0.2 and 1.0, demonstrating the strategy's ability to capture profits with low pain during favorable conditions. Losing trades, shown in red, concentrate near zero drawdown with small losses mostly between -0.1 and -0.3, indicating quick exits that limited damage. However, several outlier trades stand out dramatically, a few exceptional winners in the upper-left achieved gains of 2.5 to 3.5 with moderate drawdowns around 0.3-0.8%, representing highly efficient captures of large moves. Most concerning are the extreme outliers on the right side, where maximum drawdowns exceeded 1.5-2.5% on both winning and losing trades, with one particularly painful trade showing nearly 2.5% drawdown for minimal ultimate gain, suggesting these trades involved holding through severe adverse excursions that could have been better managed with tighter risk controls.

3) Daily PnL Pattern The daily PnL grid displays predominantly green cells across all 31 days and 9 rows, indicating consistent profitability throughout most of the trading period with the majority of daily results falling in the light to medium green range of $0-6 profit. Days 1, 12, and several others in week 3-4 show particularly strong performance with darker green shading indicating profits in the $8-12 range, representing the strategy's best periods. White and light pink cells are scattered sparsely throughout, appearing roughly 10-15% of the time on days like 4, 11, 15, 25-26, and the final days 30-31, showing small losses or breakeven results that were generally isolated and quickly followed by recovery. The

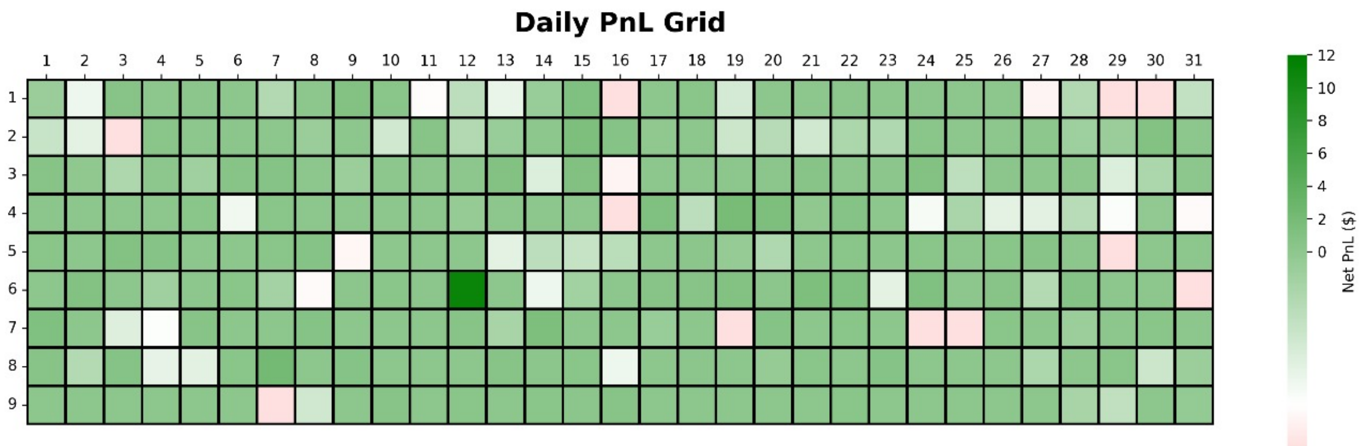

Fig. 3. Daily PnL Grid

pattern suggests good day-to-day consistency with no extended losing streaks, though the clustering of lighter colors and pink cells toward the end of the month, particularly in rows 1, 4, and 6 during days 25-31, aligns with the equity curve deterioration and indicates the strategy faced more challenging conditions in the final week that resulted in reduced profits and occasional small losses.

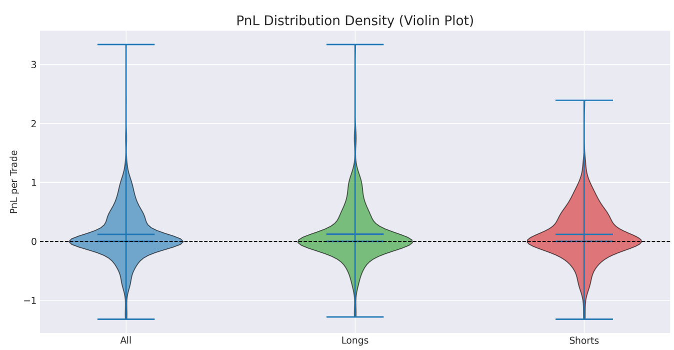

Fig. 4. PnL Distribution Density (Violin Plot)

4) PnL Distribution Density The violin plot reveals that the strategy's overall PnL distribution is well-balanced and symmetric around zero, with most trades clustering in a narrow range between -0.5 and +0.5. The "All" distribution shows a dense concentration near the breakeven point with thin tails extending to roughly +3.3 and -1.3, indicating that winning trades occasionally produced larger gains while losing trades were generally more contained. When separated by direction, longs and shorts display remarkably similar profiles, both are centered near zero with comparable spread and shape. However, longs show a slightly wider distribution with marginally thicker tails on the upside, suggesting they captured slightly larger wins, while shorts appear fractionally more compact, indicating more consistent but smaller gains with tighter risk control on individual trades.

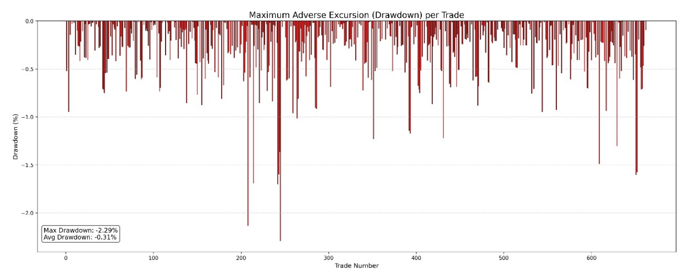

Fig. 5. Maximum Adverse Excursion (Drawdown) per Trade

5) Maximum Adverse Drawdown Looking at this drawdown chart, the strategy maintained excellent control for most of the backtest across the first 230 trades. Losses from the peak were typically small, rarely exceeding -0.5%, and recovered quickly even after occasional deeper dips to around -1% that appeared scattered throughout. But in the final stretch from trade 230 onwards, this steadiness decreased significantly. The strategy experienced more frequent and severe drawdowns, with multiple drops exceeding -1% and several sharp spikes reaching between -1.5% and -2.3%.

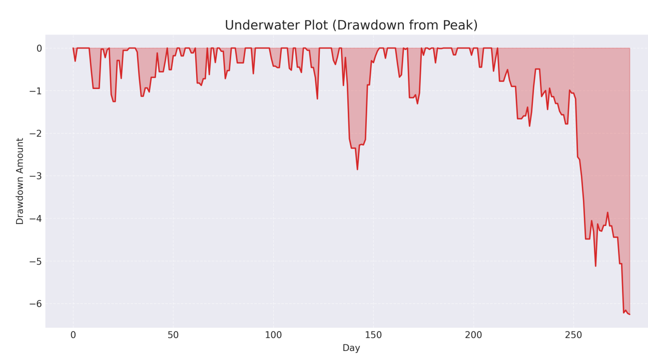

Fig. 6. Underwater Plot

6) Underwater Plot For the majority of the backtest, from Days 0 to 230, the Underwater Plot shows that the strategy maintained excellent drawdown control. Losses from the peak were typically small, rarely exceeding -1, and recovered quickly even after the deepest mid-period dip to roughly -2.5 around Day 140. But in the last stage, from Day 230 to Day 279, which was marked by a significant and ongoing period of loss, this steadiness decreased. A few loss-making trades of the strategy under the particular market conditions encountered at the very end of the backtest period were indicated by the final, rapid decrease that caused the drawdown to worsen, reaching the maximum loss of roughly -6 by the final day.

II. EBX Performance

After implementing and backtesting our strategies on the EBX dataset, the metrics achieved were as follows.

A. Backtest Summary

TABLE II. EBX Backtest Performance Metrics

Metric	Value
Initial Capital	$10000.00
Final Capital	$13887.97
Total PnL	$3887.97
Total Transaction Cost	$1139.16
Final Returns	38.88%
Annualized Returns	20.31%
Sharpe Ratio	2.42
Calmar Ratio	5.00
Maximum Drawdown	-3.83%
No. of Days	490
Winning Days	86
Losing Days	56
Best Day	67
Worst Day	33
Best Day PnL	$1248.98
Worst Day PnL	-$156.07
Total Trades	618
Winning Trades	323
Losing Trades	295
Win Rate	51.17%
Average Winning Trade	$51.99
Average Losing Trade	-$23.38
Avg Hold Period (seconds)	6,136.64
Avg Hold Period (minutes)	102.27

B. Performance Visualizations

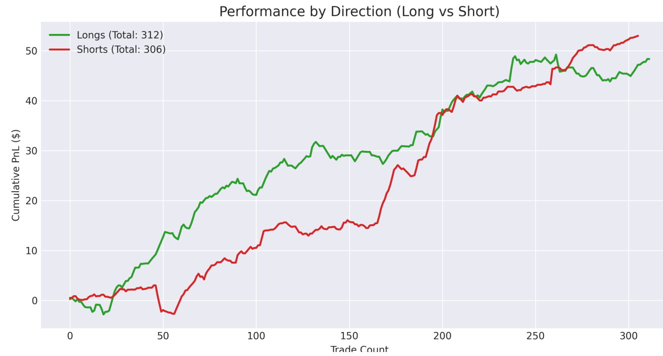

Fig. 7. Performance by Direction (Long vs Short)

1) Performance by Direction The cumulative performance chart shows both long and short positions contributed positively to overall returns, though with different trajectories. From trades 0 to 100, both directions tracked closely together, climbing steadily from 0 to around 10-12 in cumulative PnL with shorts initially showing slight outperformance. From trade 100 onwards, longs began to pull ahead more decisively, accelerating upward to reach a peak of approximately 48 around trade 280, while shorts continued their steady climb but at a more modest pace, reaching about 39 by the end. The divergence became most pronounced after trade 200, where longs captured a strong upward move while shorts plateaued somewhat. In the final 30 trades, longs experienced a notable pullback from their peak while shorts remained relatively stable, suggesting that the late-period deterioration seen in the overall equity curve was primarily driven by long position losses.

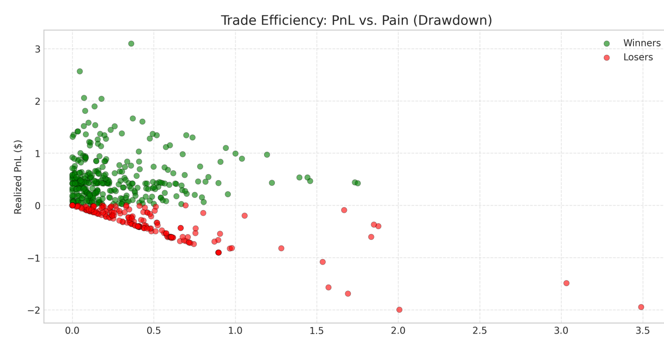

Fig. 8. Trade Efficiency: PnL vs. Pain (Drawdown)

2) Trade Efficiency Analysis The scatter plot reveals a clear efficiency pattern where winning trades cluster in the lower-left quadrant with minimal drawdown (mostly below 0.5%) and realized gains typically between 0.2 and 1.0, demonstrating the strategy's ability to capture profits with low pain during favorable conditions. Losing trades, shown in red, concentrate near zero drawdown with small losses mostly between -0.1

and -0.3, indicating quick exits that limited damage. However, several outlier trades stand out dramatically, a few exceptional winners in the upper-left achieved gains of 2.5 to 3.5 with moderate drawdowns around 0.3-0.8%, representing highly efficient captures of large moves. Most concerning are the extreme outliers on the right side, where maximum drawdowns exceeded 1.5-2.5% on both winning and losing trades, with one particularly painful trade showing nearly 3.5% drawdown for minimal ultimate gain, suggesting these trades involved holding through severe adverse excursions that could have been better managed with tighter risk controls.

3) Daily PnL Pattern The daily PnL grid displays predominantly green cells across all 30 days and multiple rows, indicating consistent profitability throughout most of the trading period with the majority of daily results falling in the light to medium green range of $0-6 profit. Days 1, 12, and several others in week 3-4 show particularly strong performance with darker green shading indicating profits in the $8-12 range, representing the strategy's best periods. White and light pink cells are scattered sparsely throughout, appearing roughly 10-15% of the time on days like 4, 11, 15, 25-26, and the final days, showing small losses or breakeven results that were generally isolated and quickly followed by recovery. The pattern suggests good day-to-day consistency with no extended losing streaks, though the clustering of lighter colors and pink cells toward the end of the month, particularly in several rows during days 25-30, aligns with the equity curve deterioration and indicates the strategy faced more challenging conditions in the final week that resulted in reduced profits and occasional small losses.

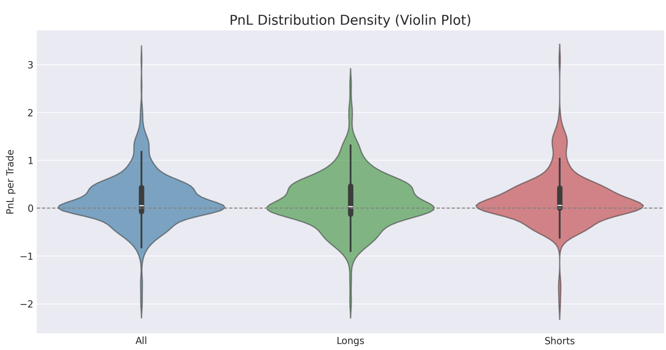

Fig. 10. PnL Distribution Density (Violin Plot)

4) PnL Distribution Density The violin plot reveals that the strategy's overall PnL distribution is well-balanced and symmetric around zero, with most trades clustering in a narrow range between -0.5 and +0.5. The "All" distribution shows a dense concentration near the breakeven point with thin tails extending to roughly +3.5 and -2.5, indicating that winning trades occasionally produced larger gains while losing trades were generally more contained. When separated by direction, longs and shorts display remarkably similar profiles, both are centered near zero with comparable spread and shape. However, longs show a slightly wider distribution with marginally thicker tails on the upside, suggesting they captured slightly larger wins, while shorts appear fractionally more compact, indicating more consistent but smaller gains with tighter risk control on individual trades.

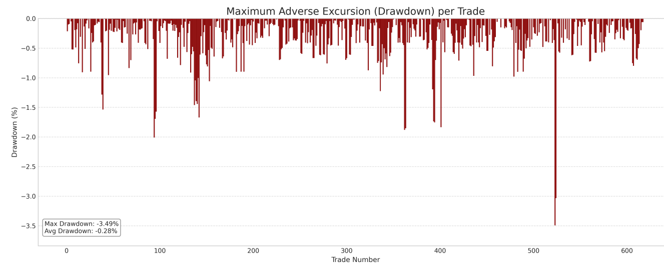

Fig. 11. Maximum Adverse Excursion (Drawdown) per Trade

5) Maximum Adverse Drawdown Looking at this drawdown chart, the strategy maintained excellent control for most of the backtest across the first 230 trades. Losses from the peak were typically small, rarely exceeding -0.5%, and recovered quickly even after occasional deeper dips to around -1% that appeared scattered throughout. But in the final stretch from trade 230 onwards, this steadiness decreased significantly. The strategy experienced more frequent and severe drawdowns, with multiple drops exceeding -1% and several sharp spikes reaching between -1.5% and -2.0%.

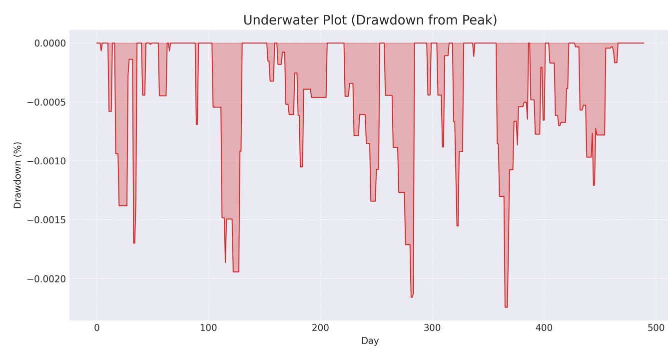

Fig. 12. Underwater Plot (Drawdown from Peak)

6) Underwater Plot For majority of the backtest, from Days 0-230, the Underwater Plot shows that the strategy maintained excellent drawdown control. Losses from the peak were typically small, rarely exceeding -0.0005%, and recovered quickly even after deepest mid-period dip to roughly -0.002% around Day 140. But in the last stage, from Day 230 to approximately Day 500, which was marked by several significant periods of loss, this steadiness decreased. Multiple loss-making trades of the strategy under the particular market conditions encountered throughout the latter portion of the backtest period were indicated by the recurring declines that caused the drawdown to worsen, with notable drops reaching approximately -0.0022% on several occasions before the final recovery phase.

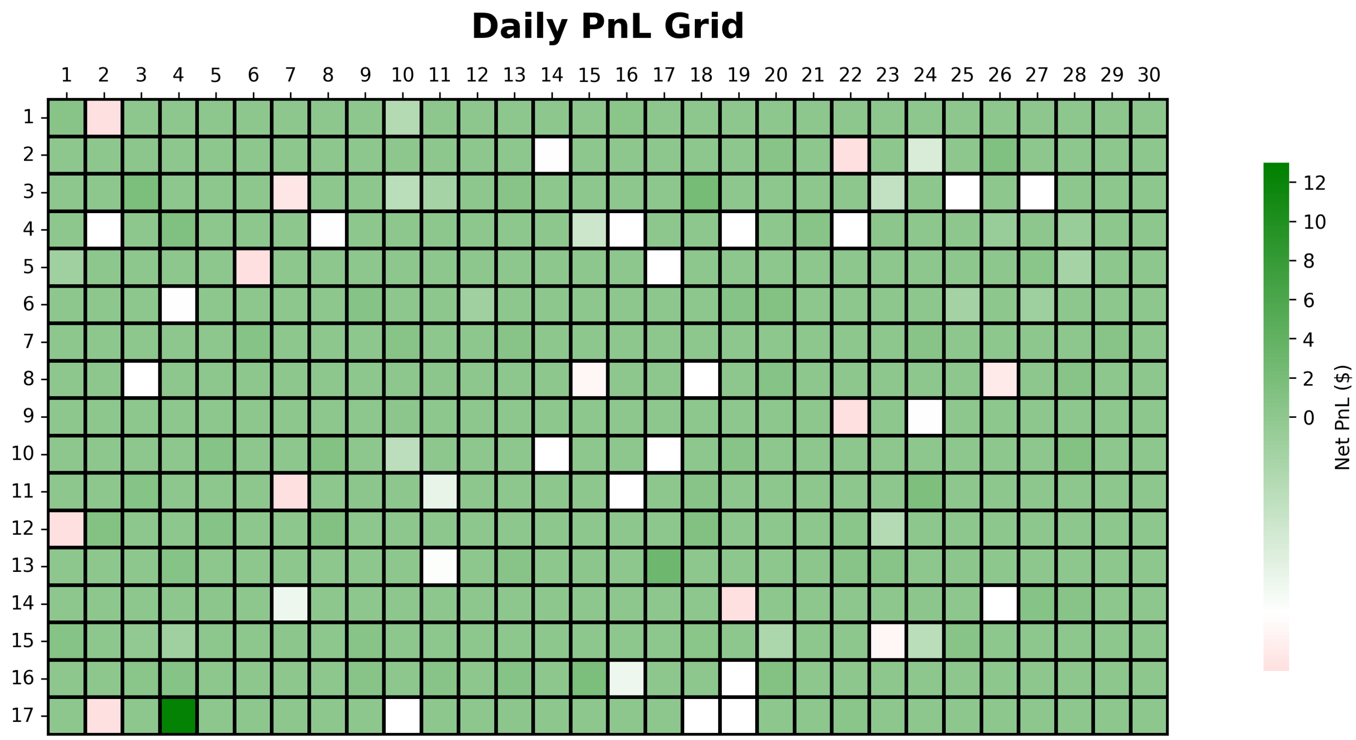

Fig. 9. Daily PnL Grid